



Complete Paper Solution

EGD – Summer 2022

Engineering Graphics and

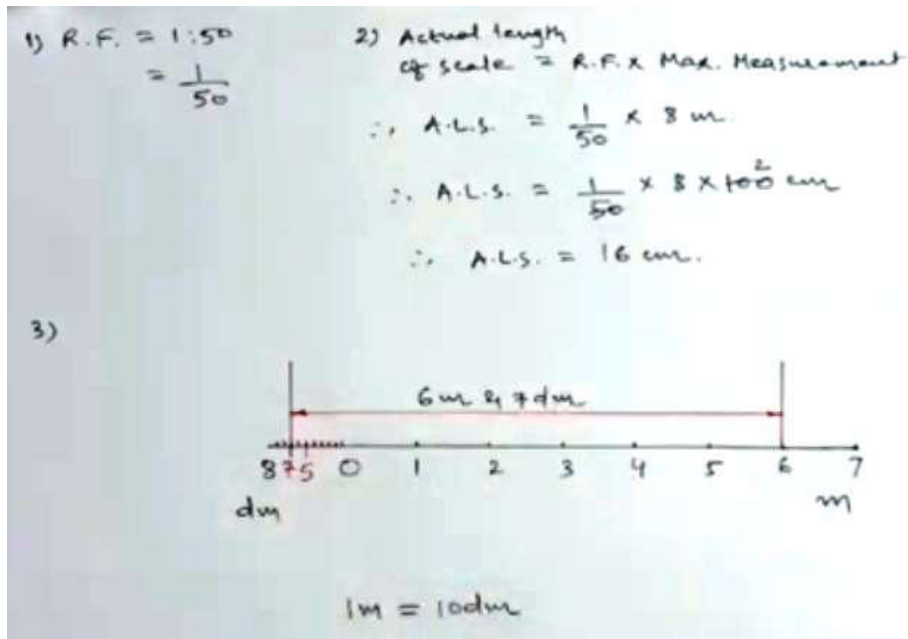
Design (GTU)

Q.1 (a) Explain Representative Fraction. Construct a plain scale of R.F. = 1:50 to show meters and decimeters and long enough to measure 8 meters. Indicate 6.7 m distance on it. (**03 Marks**)

Answer:

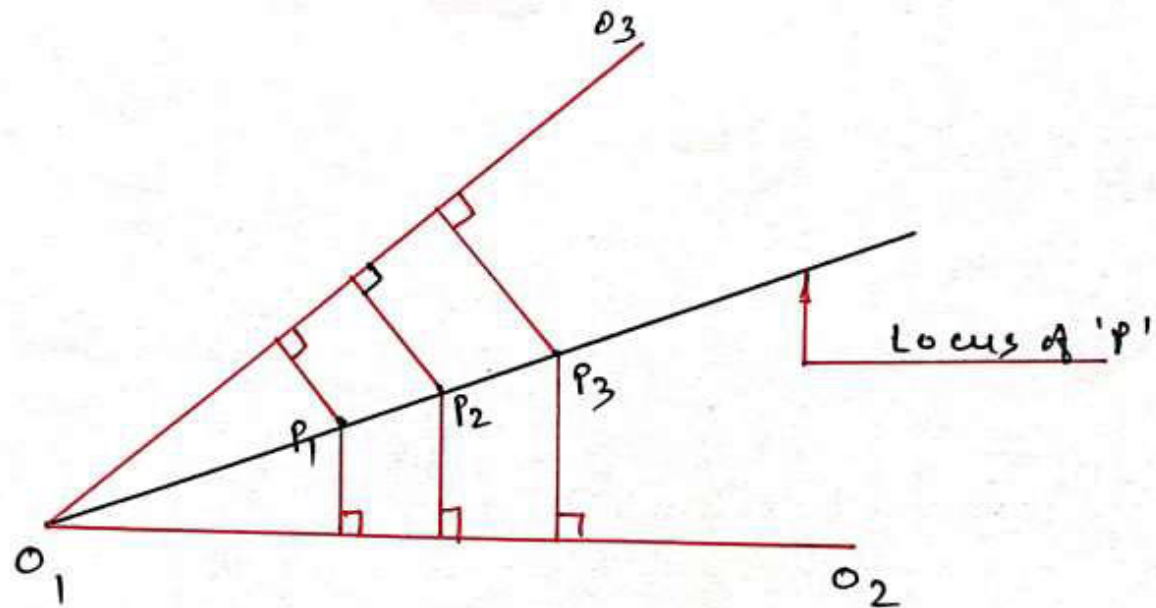
Representative Fraction is the ratio of size of object in paper to actual size of object.

$$\text{Representative Fraction (R.F.)} = \frac{\text{Size of object in paper}}{\text{Actual size of object}}$$



[Video Link: https://youtu.be/V1iRKikieZA](https://youtu.be/V1iRKikieZA)

Q.1 (b) Find the locus of a point P, moving in a plane, keeping its distances equal from two non-parallel fixed straight lines O_1O_2 and O_1O_3 . (04 Marks)





Q. 1 (c) Crank OA rotates about 'O'. (Slide along 'OC', rod BD is fixed to connecting rod AC). Draw locus of point 'D' for one complete rotation of crank OA. (Refer figure 1) (**07 Marks**)

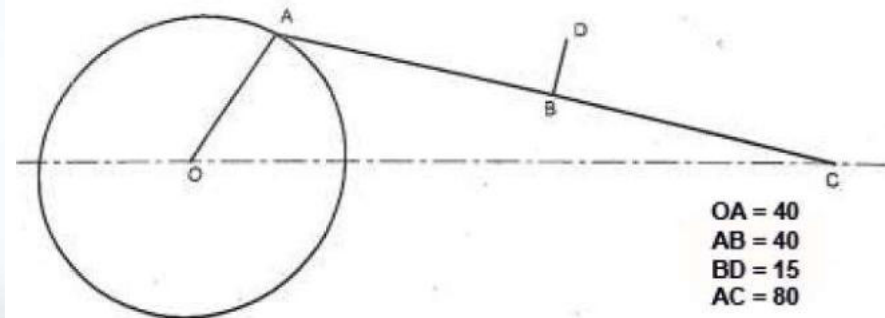
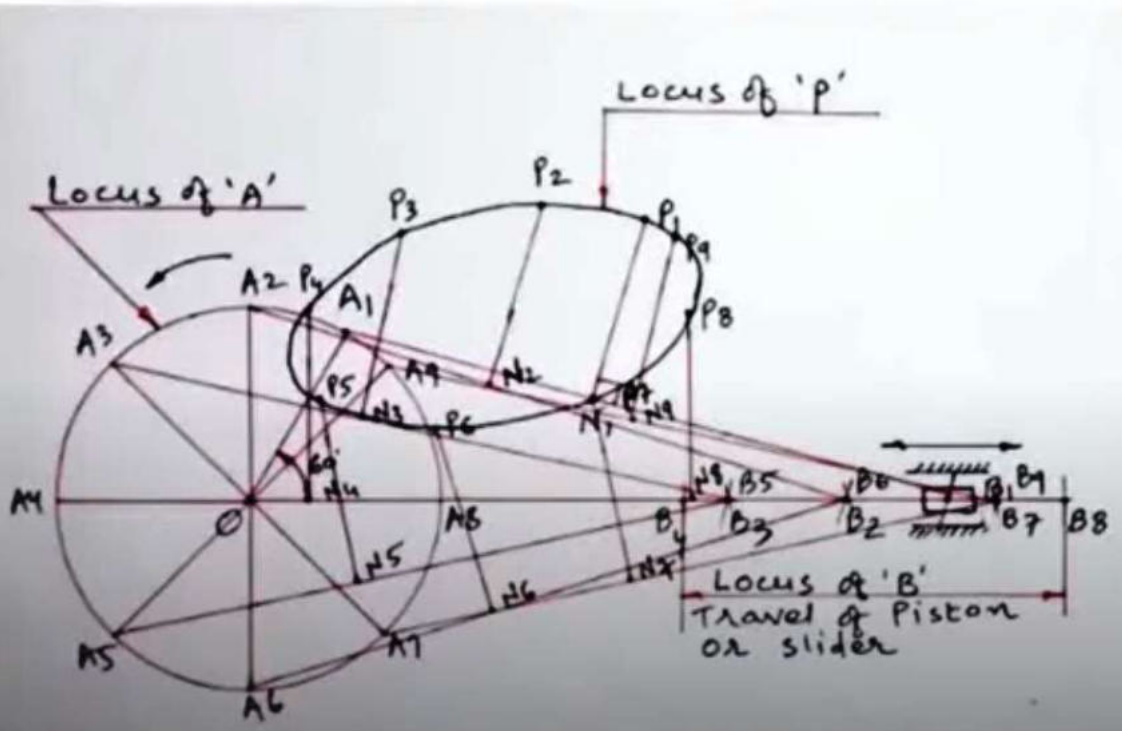
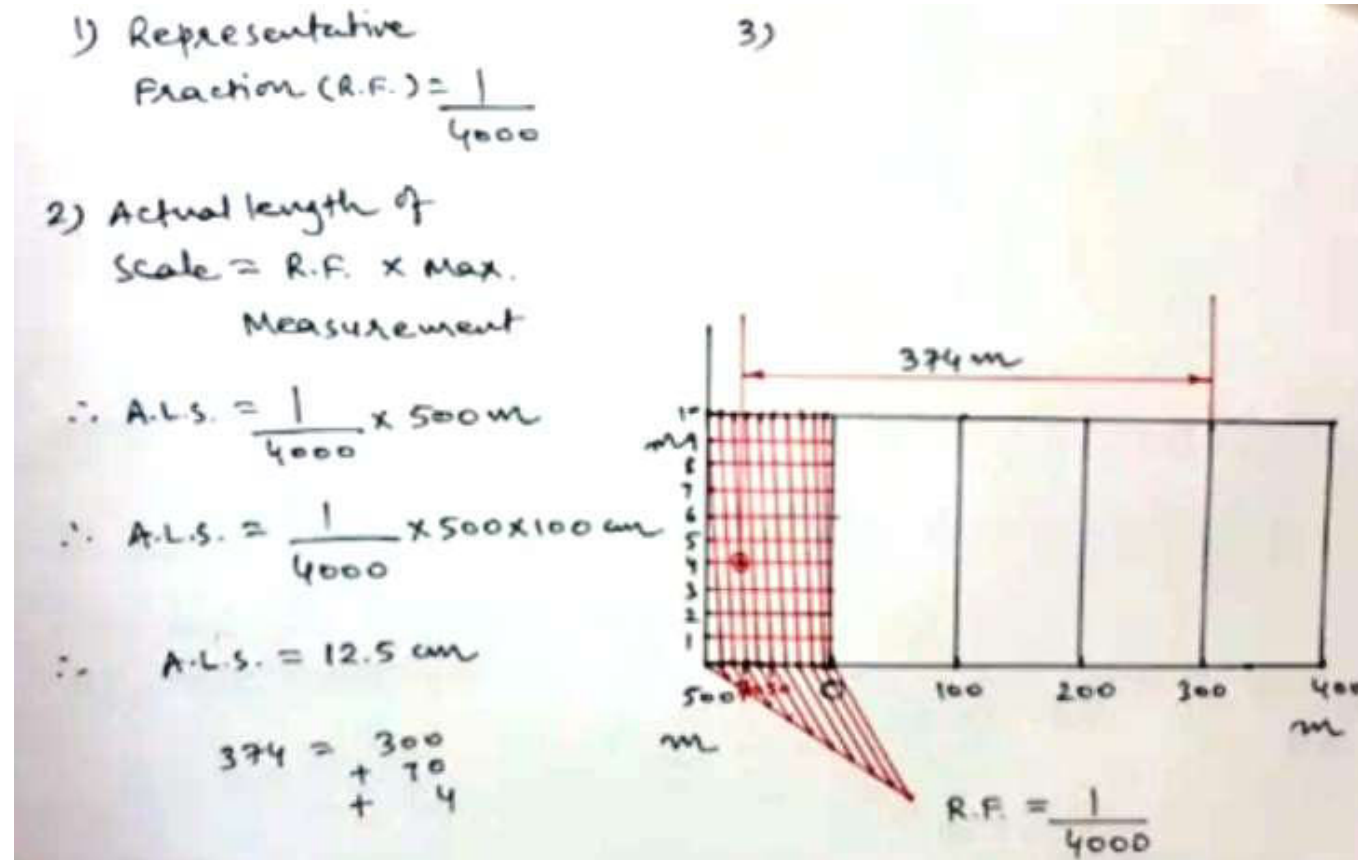


Figure 1

Video Link: <https://youtu.be/L1YggM863rw>

Q.2 (a) Construct a diagonal scale of RF = 1/4000 to show 374 meters and long enough to measure up to 500 meters. (03 Marks)

GTU Summer 2018



Video Link: <https://youtu.be/AwqpFQHoDF8>



Q. 2 (b) The distance between Delhi and Agra is 200 km. In a railway map it is represented by a line 5 cm long. Find its R.F. Draw a diagonal scale to show single km. And maximum 600 km. Indicate on it the distance of 322 km. (04 Marks)

Representative Fraction
(R.F.) = $\frac{\text{size of object in paper}}{\text{Actual size of object}}$

$$= \frac{5 \text{ cm}}{200 \text{ km}}$$

$$= \frac{5 \text{ cm}}{200 \times 1000 \times 100 \text{ cm}}$$

$$\text{R.F.} = \frac{4}{40,00,000}$$

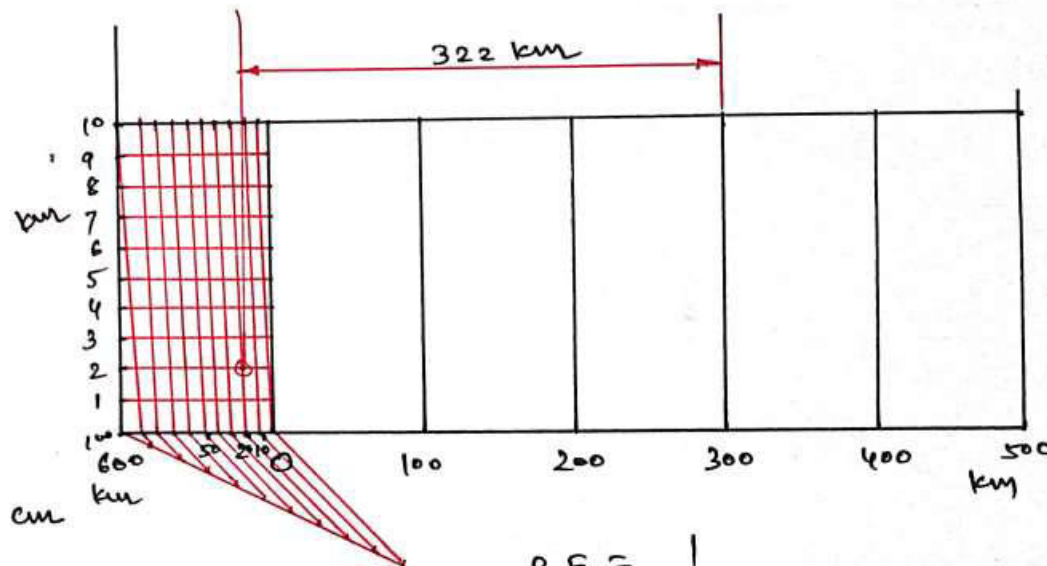
A.L.S. = R.F. $\times$ Max. Measurement

$$= \frac{1}{40,00,000} \times 600 \text{ km}$$

$$= \frac{1}{40,00,000} \times 600 \times 1000 \times 100 \text{ cm}$$

$$= 15 \text{ cm}$$

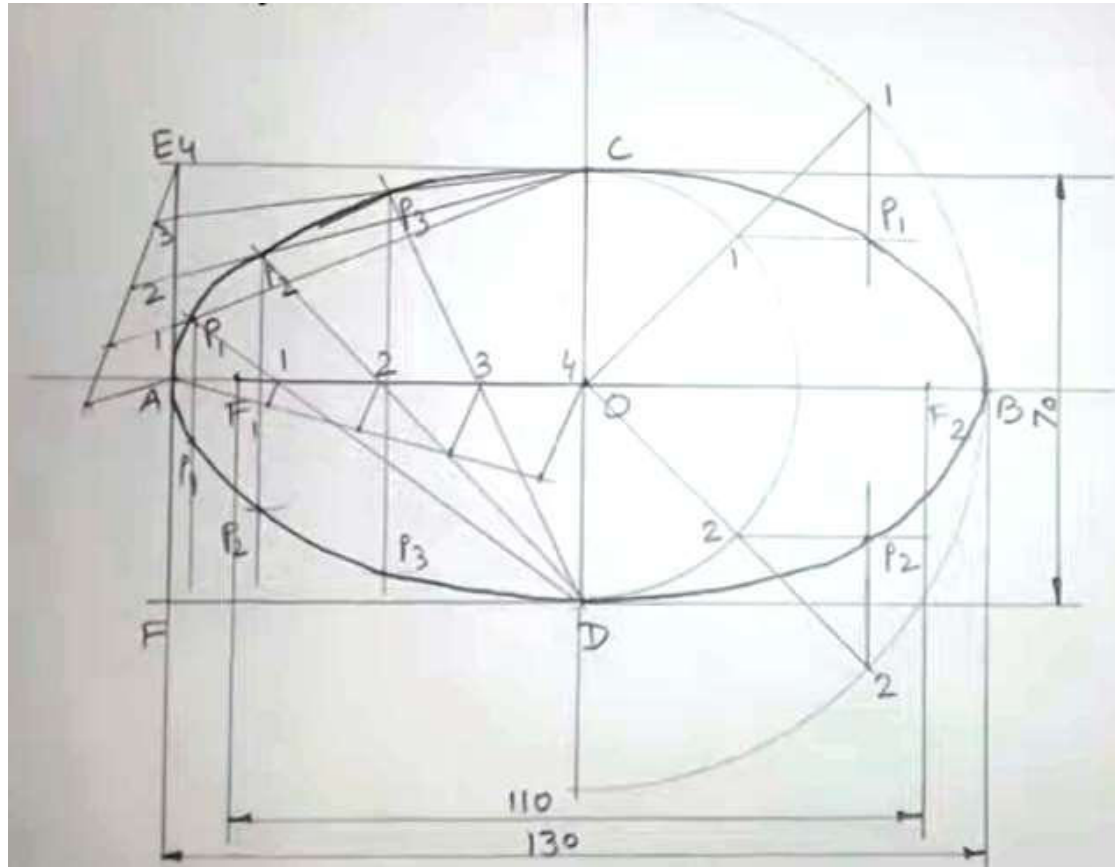
$$\begin{array}{r} 322 = 300 \\ \quad 20 \\ \quad \quad 2 \\ \hline 322 \end{array}$$



$$\text{R.F.} = \frac{1}{40,00,000}$$

Q. 2 (c) The foci of an ellipse are 110 mm apart. The minor axis is 70 mm long. Determine the length of major axis and draw half ellipse by rectangle method and other half by concentric circle method. (07 Marks)

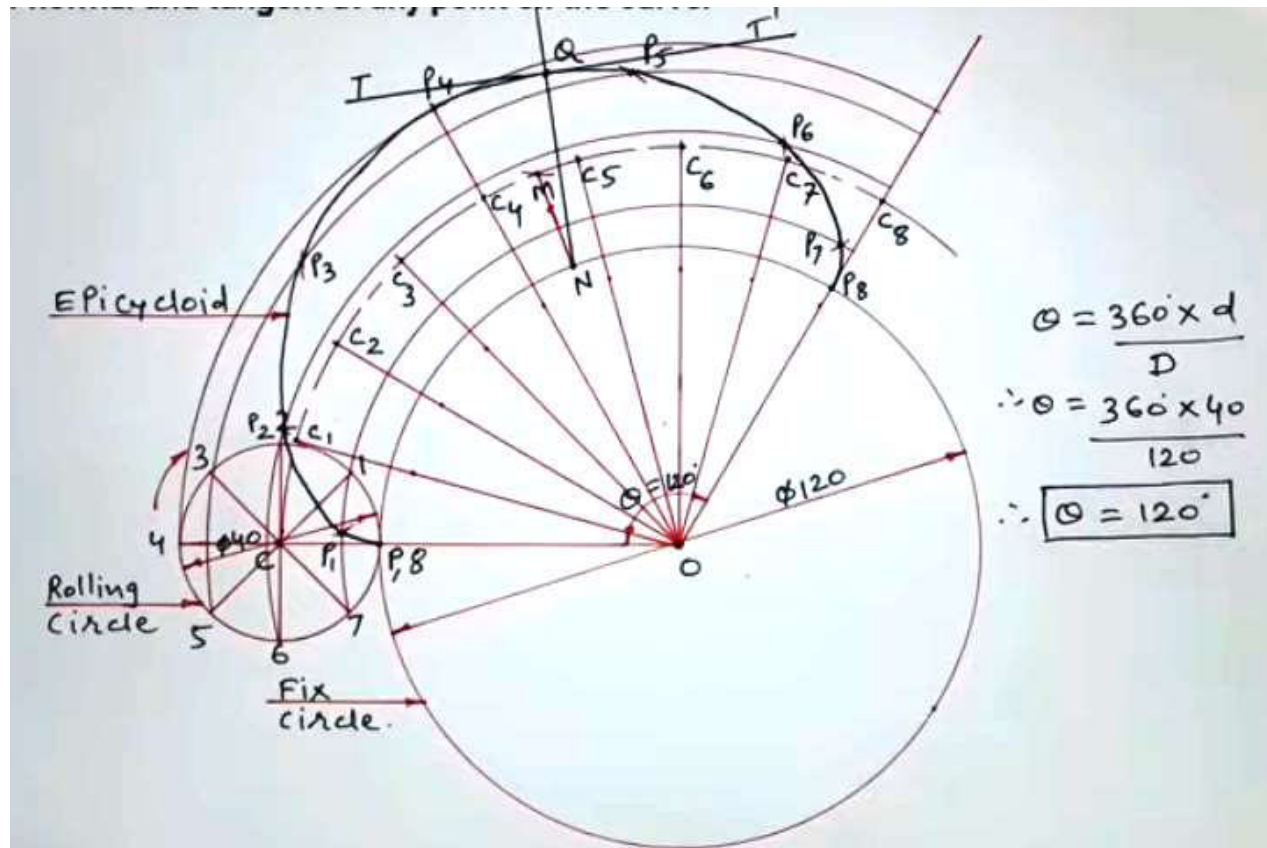
GTU Winter 2015



[Video Link: https://youtu.be/t4kCkExOmGc](https://youtu.be/t4kCkExOmGc)



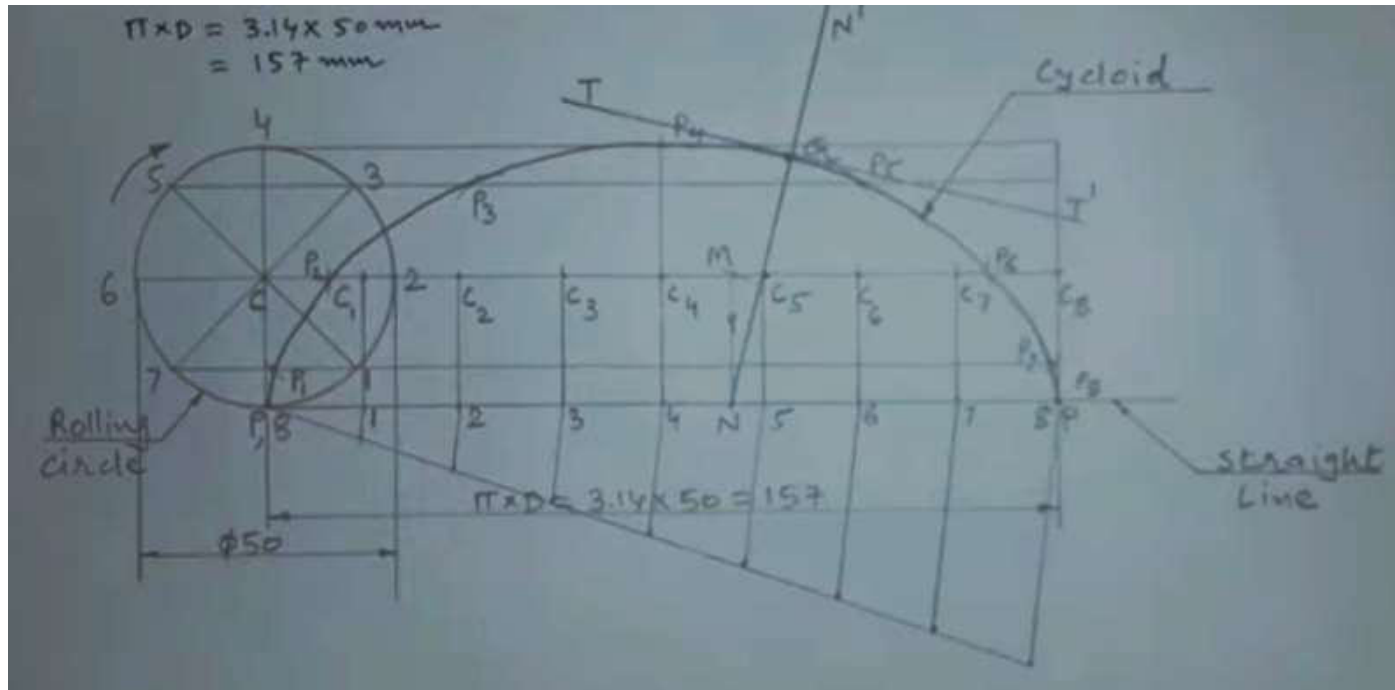
Q. 2 (c) OR A circle of 50 mm diameter rolls on the circumference of another circle of 150 mm diameter and outside it. Draw the locus of the point P on the circumference of the rolling circle for one complete revolution of it. Take initial position of point P at the contact point between two circles. Name the curve. **(07 Marks)**



Video Link: <https://youtu.be/INNwE4H6-FU>



Q.3 (a) Draw a cycloid for a rolling circle of 50 mm diameter rolling along a straight line without slipping. Take initial position of the tracing point at the bottom of the vertical centre line of the rolling circle.. (03 Marks)



Video Link: <https://youtu.be/ukuCJLdZckw>



Q. 3 (b) Define the following terminology with respect to conics and also show them in a diagram:

(i) Focus, (ii) Directrix, (iii) Axis, (iv) Vertex (**04 Marks**)

1. Focus (F):

The focus (or foci-plural) of a conic section is the point (or points) from which the distance to each point on the graph of the conic section is constant as defined by the specific conic section's equation.

2. Directrix (D-D1):

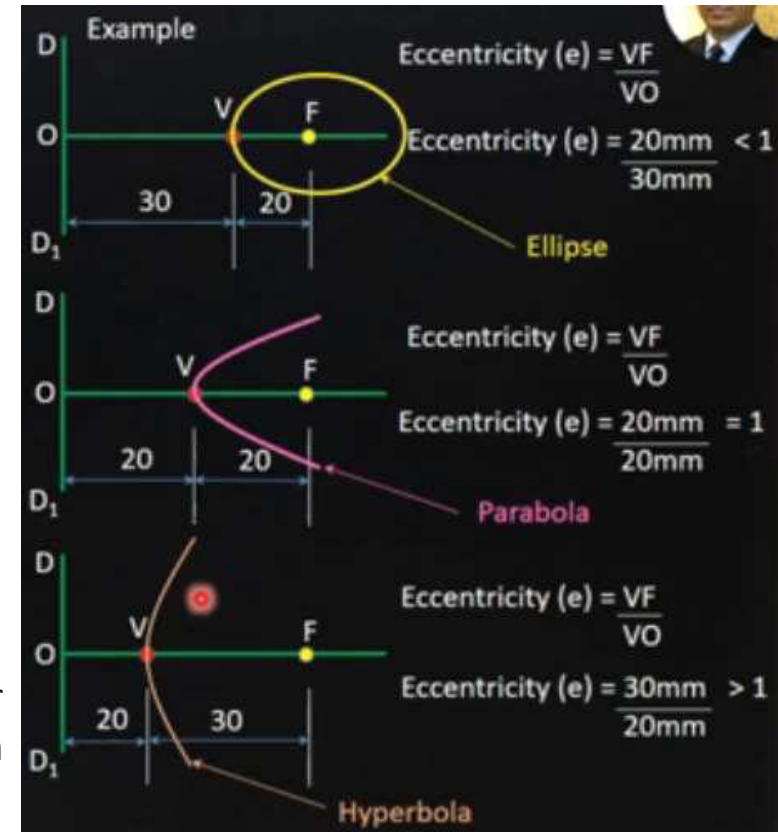
A straight line the distance to which from any point of a conic section is in fixed ratio to the distance from the same point to a focus.

3. Axis (Line through point O):

The axes of the conics are the axes of symmetry for the figure means that the two sides of the graph on either side of the axis look like mirror images of each other.

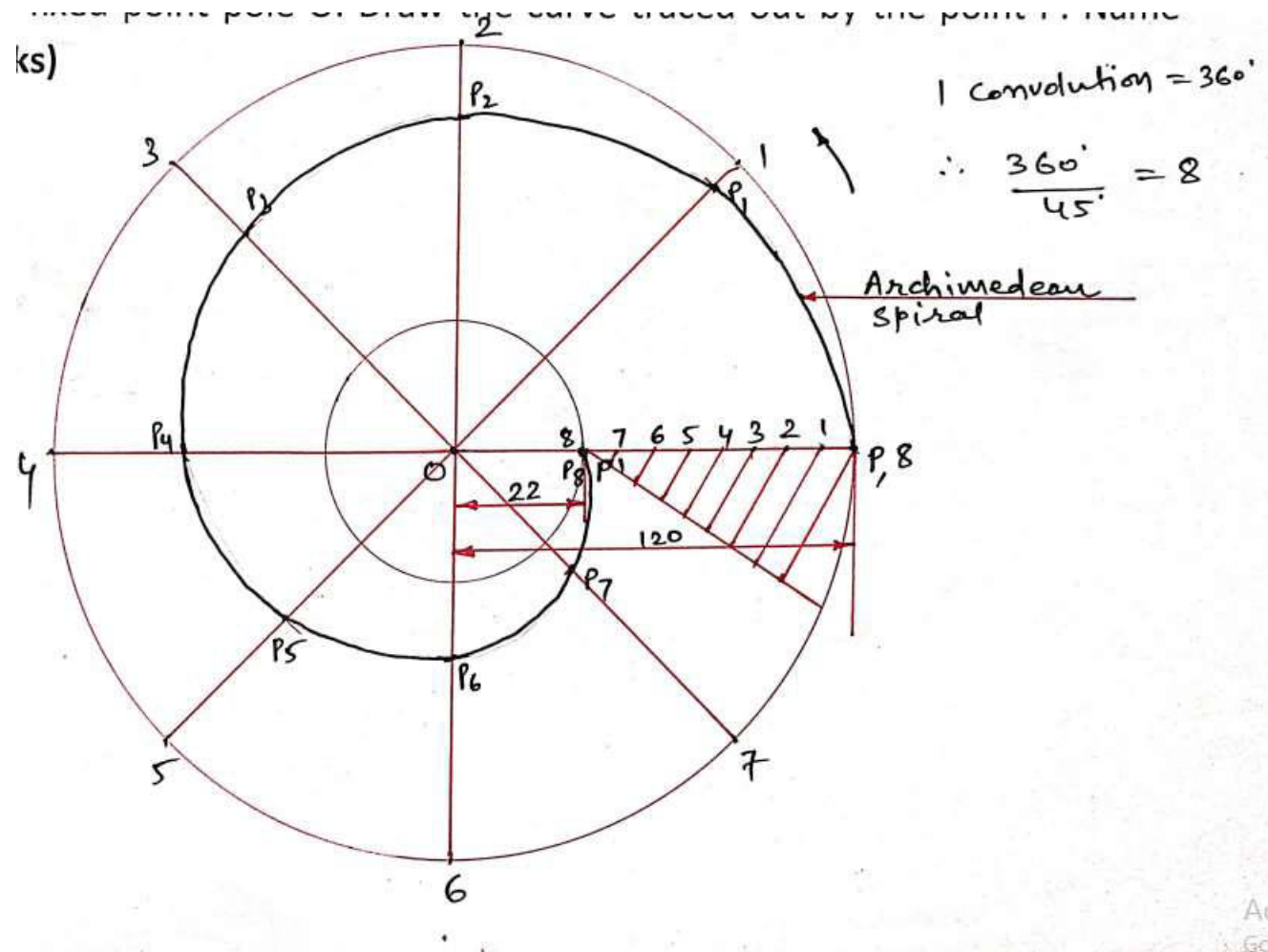
4. Vertex (V):

Vertex is a point that lies between point on directrix and focus. For parabola, the point vertex is halfway between the focus and the directrix. For an ellipse, vertex is nearer to focus and for hyperbola, vertex is nearer to directrix.

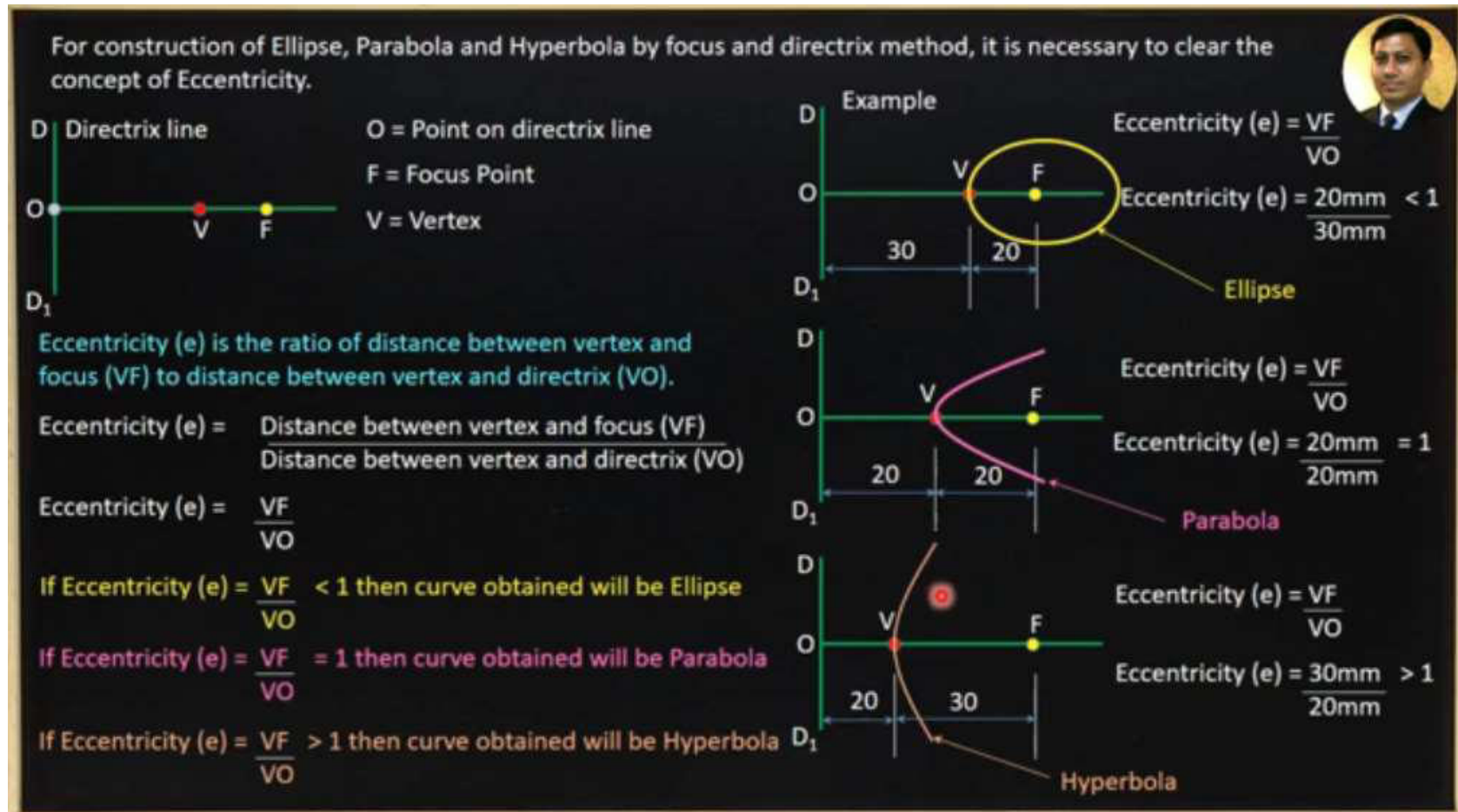




Q. 3 (c) A point P is 120 mm away from the fixed point pole O. A point P moves towards pole O and reaches the position P' in one convolution where OP' is 22 mm. The point P moves in such a way that its movement towards fixed point O, being uniform with its movement around fixed point pole O. Draw the curve traced out by the point P. Name the curve. **(07 Marks)**



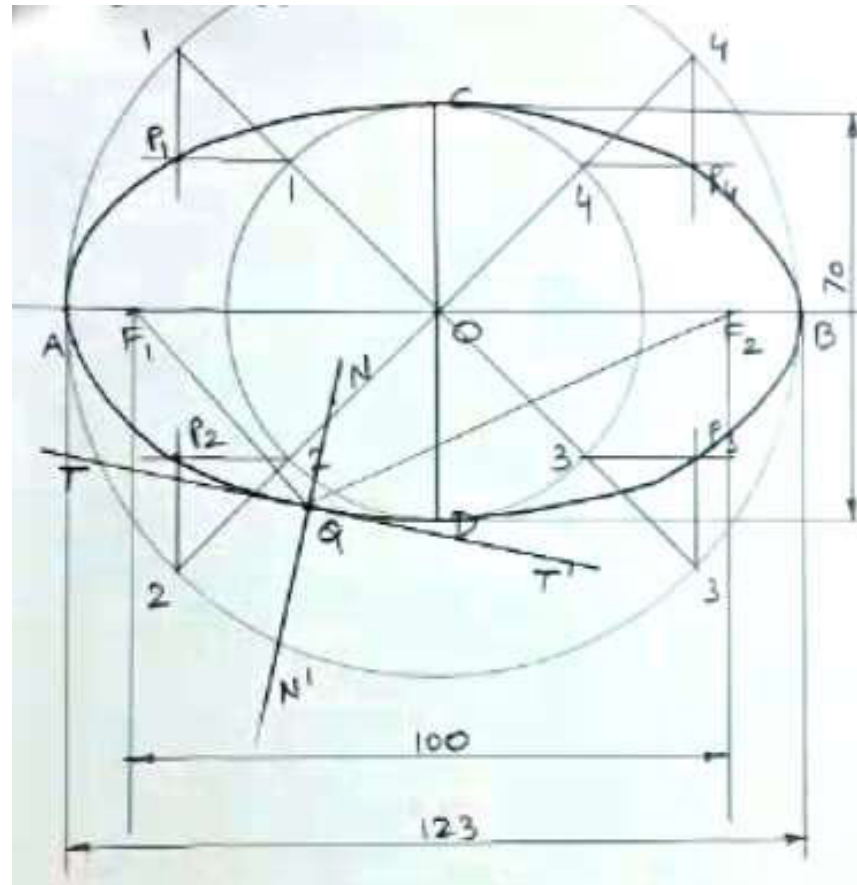
Q.3 (a) OR Define eccentricity ratio. Write the values of eccentricity for (i) ellipse, (ii) parabola, and (iii) hyperbola.
(03 Marks)



Video Link: <https://youtu.be/oAaRbw5y16w>



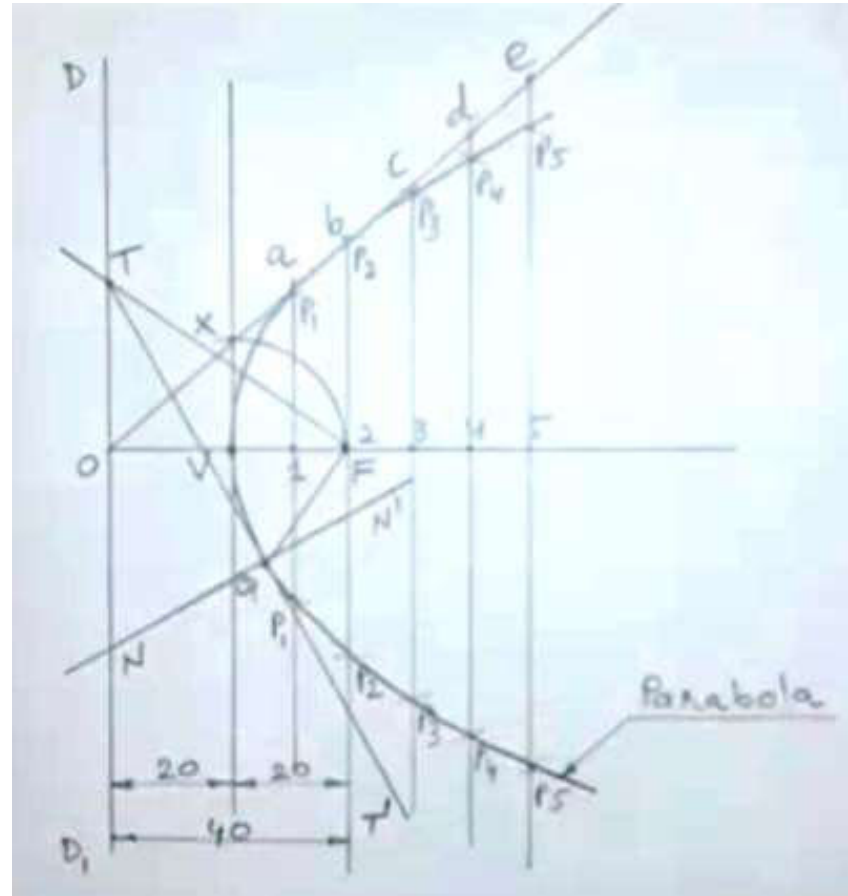
Q. 3 (b) OR Construct an ellipse foci of an ellipse is 100 mm and minor axis 70 mm by using concentric circle method.. (04 Marks)



Video Link: https://youtu.be/nwFewok1_O4



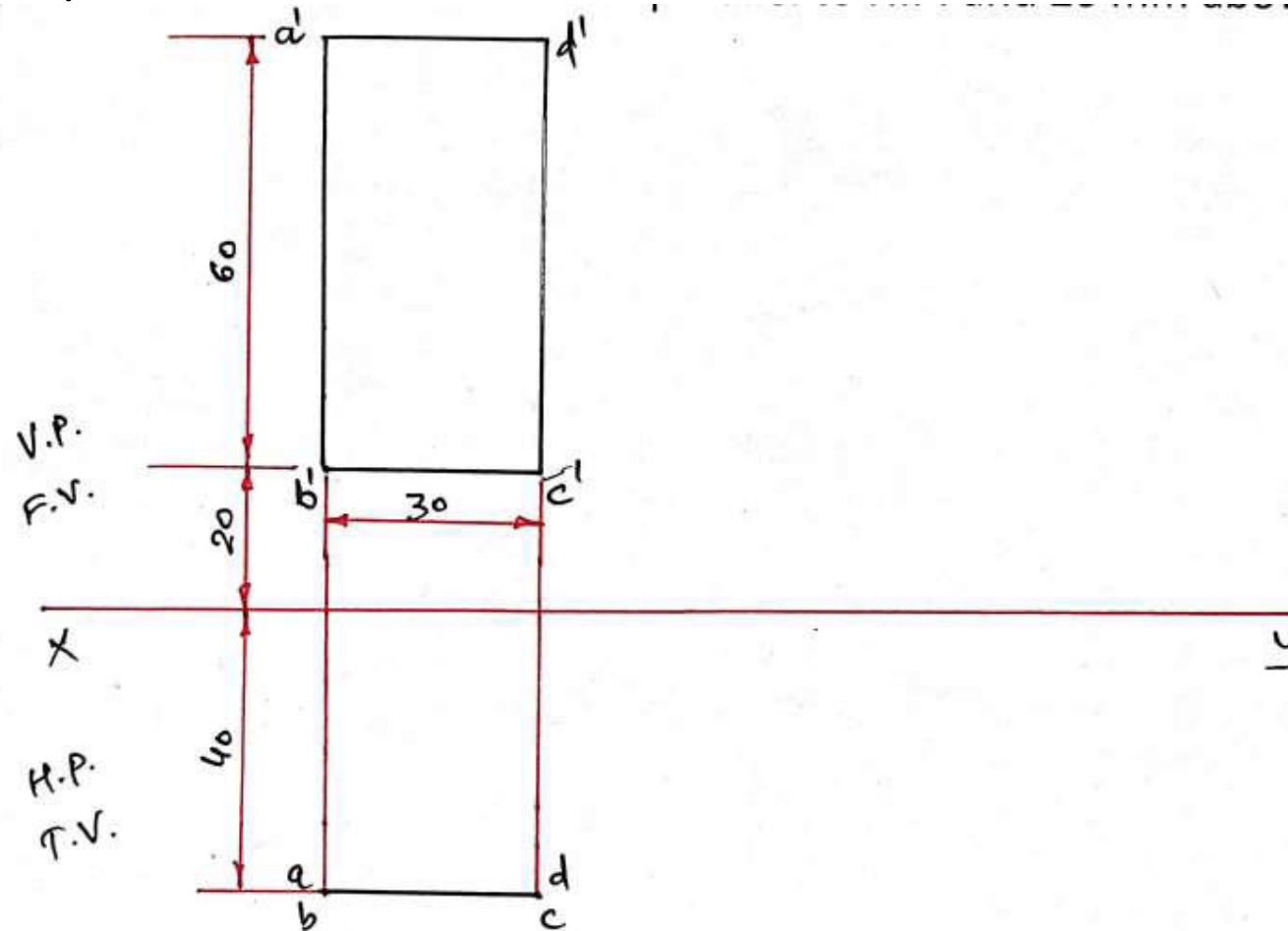
Q. 3 (c) OR Construct a parabola, when the distance of the focus from the directrix is 40 mm. Also draw a tangent and normal at a point on it 30 mm from F. (07 Marks)



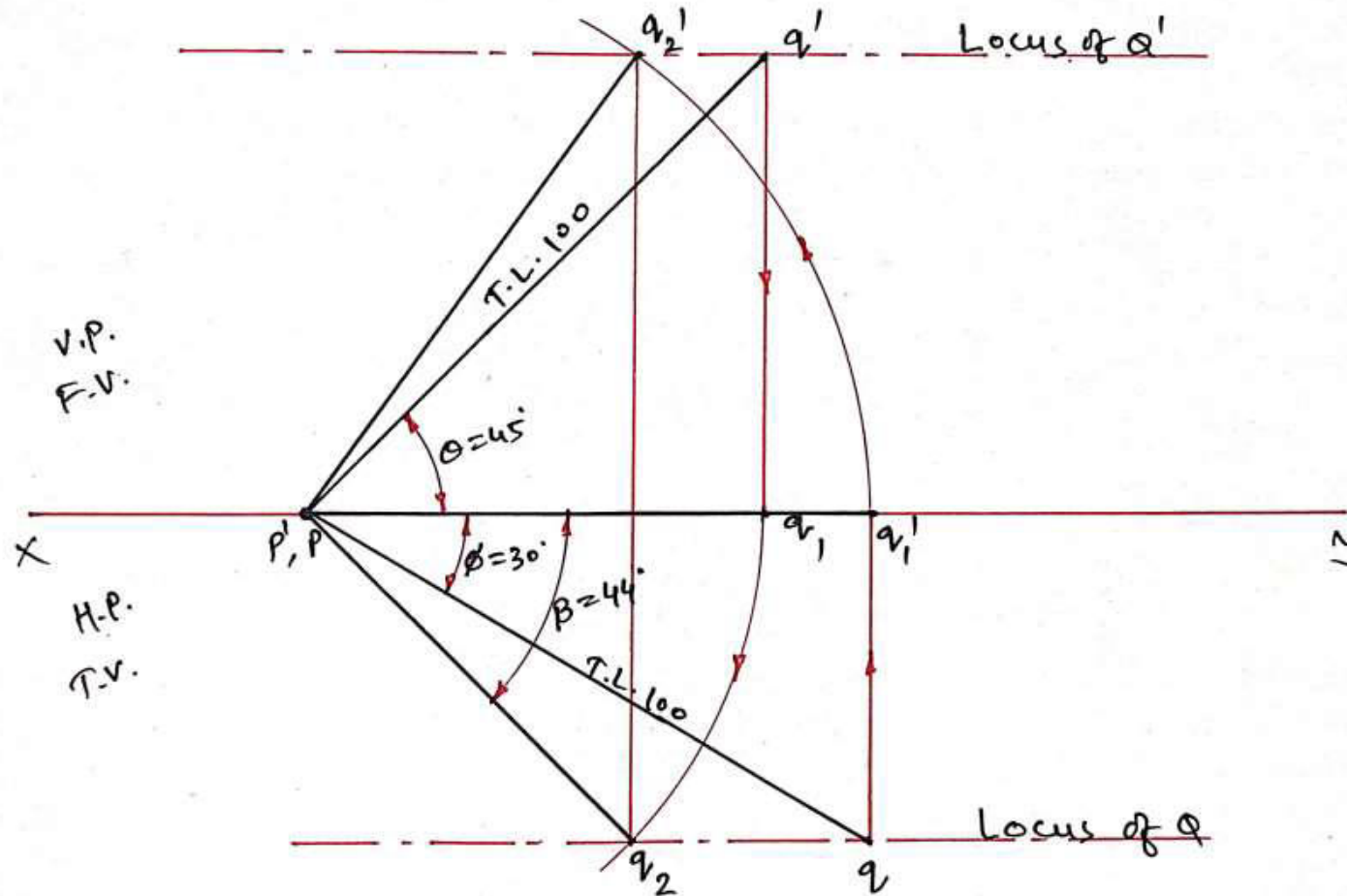
Video Link: https://youtu.be/O1PA_FXzjKs



Q.4 (a) A rectangular plane ABCD having 60 mm X 30 mm size is parallel to V.P., and perpendicular to H.P., and P.P. Draw the projections of the rectangle when it is 40 mm in front of V.P., and one of the smaller sides is parallel to H.P. and 20 mm above it. (03 Marks)

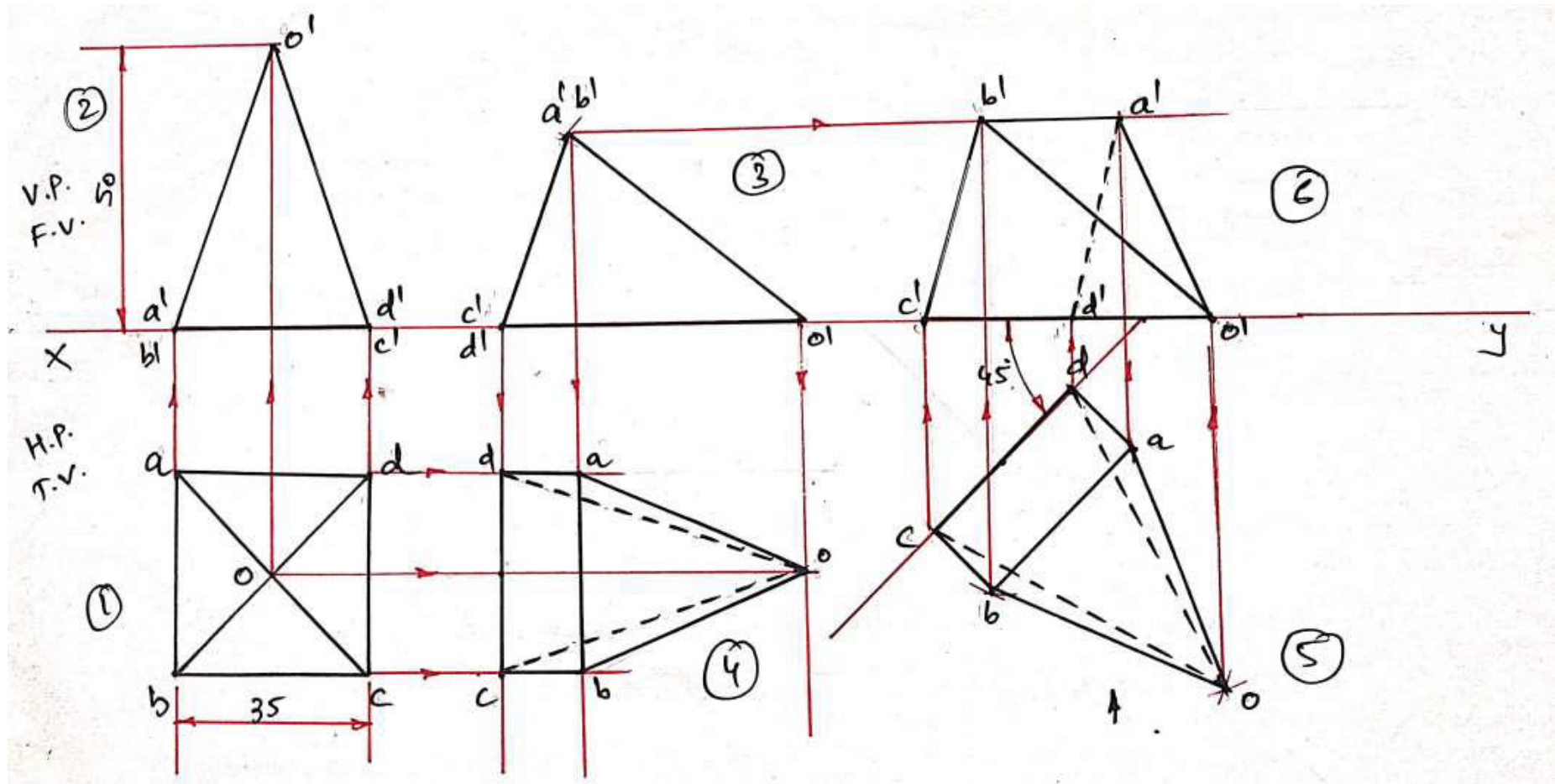


Q.4 (b) A line PQ 100 mm long is inclined at an angle of 45° to H.P. and 30° to V.P. One of its end point 'P' is in H.P. as well as VP. Determine its apparent inclination with V.P.. (04 Marks)

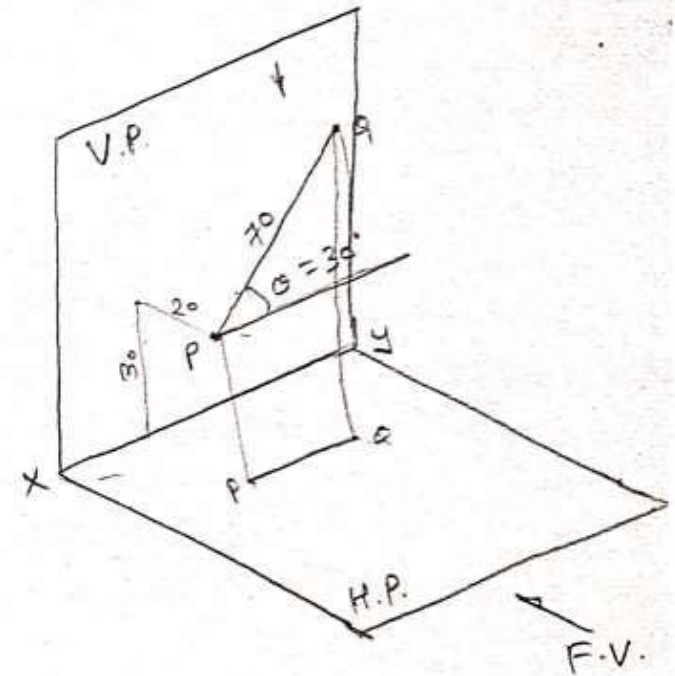
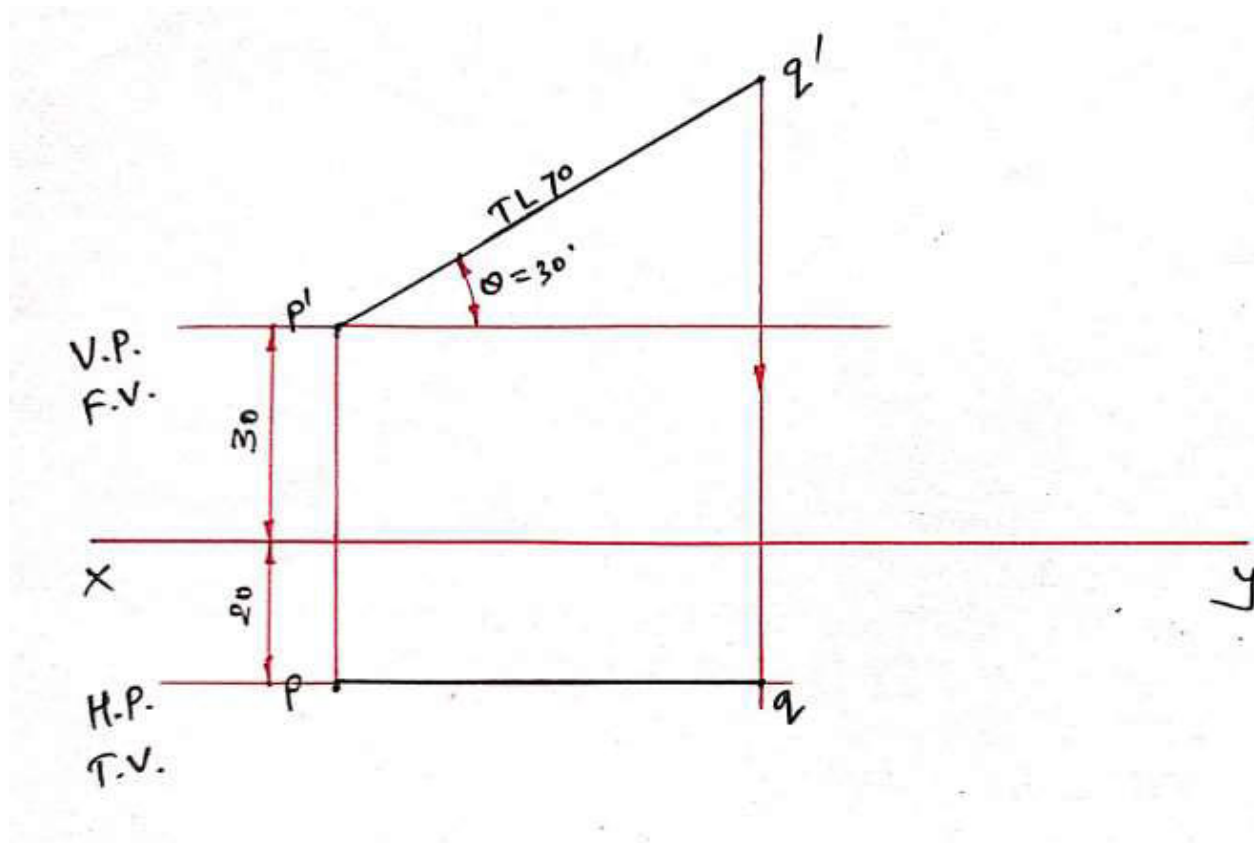




Q. 4 (c) A square pyramid, side of base 35 mm and axis length 50 mm is lying on the H.P. on one of its triangular faces. Draw the projections of the pyramid when the base edge contained by the triangular face on H.P. makes an angle of 45° on the V.P. keeping apex of the pyramid towards the observer. (07 Marks)



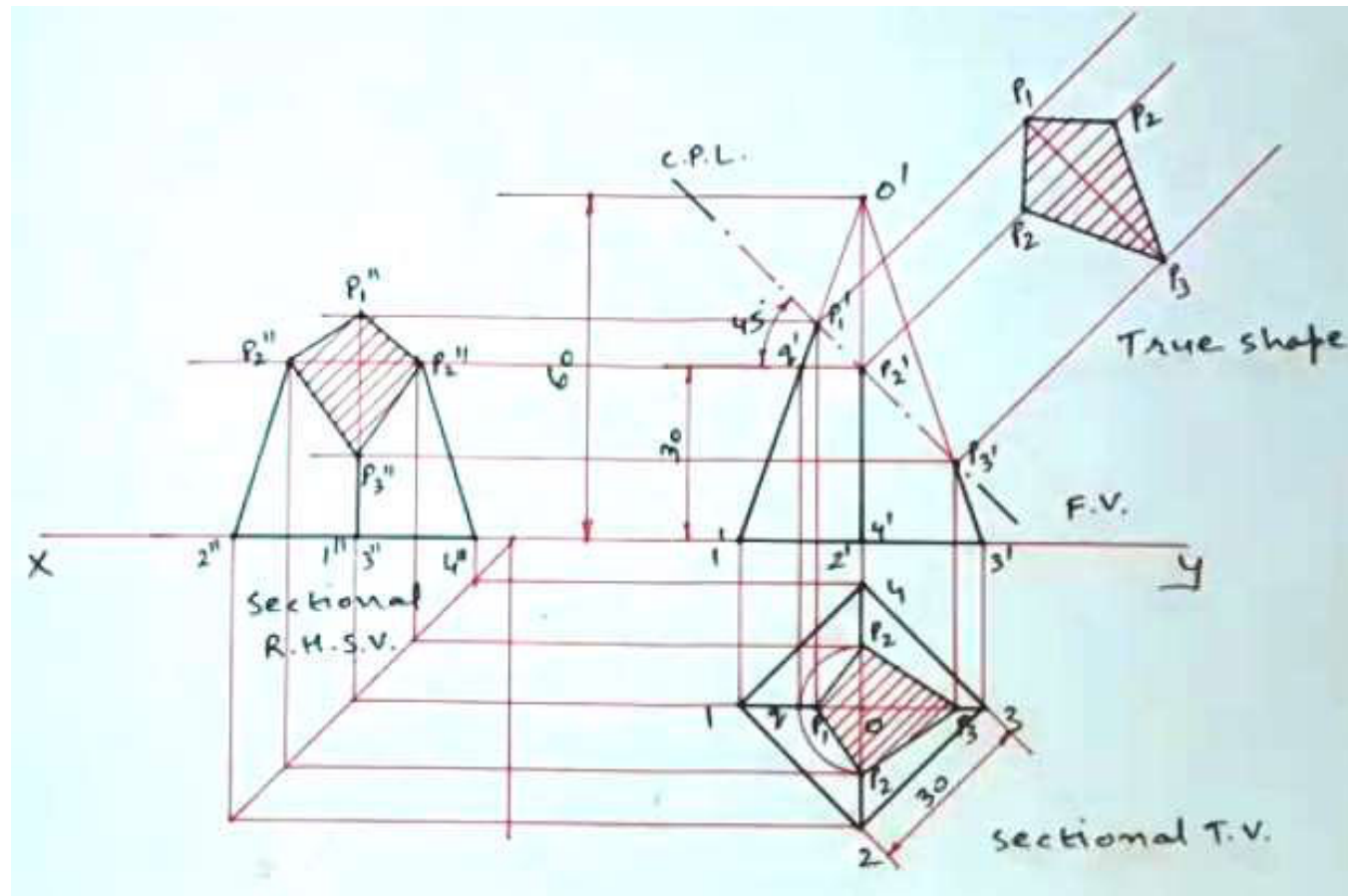
Q.4 (a) OR A line PQ 70 mm long is parallel to V.P. and 30° inclined to H.P. The end P is 30 mm above H.P. and 20 mm in front of V.P. Draw the projections. **(03 Marks)**

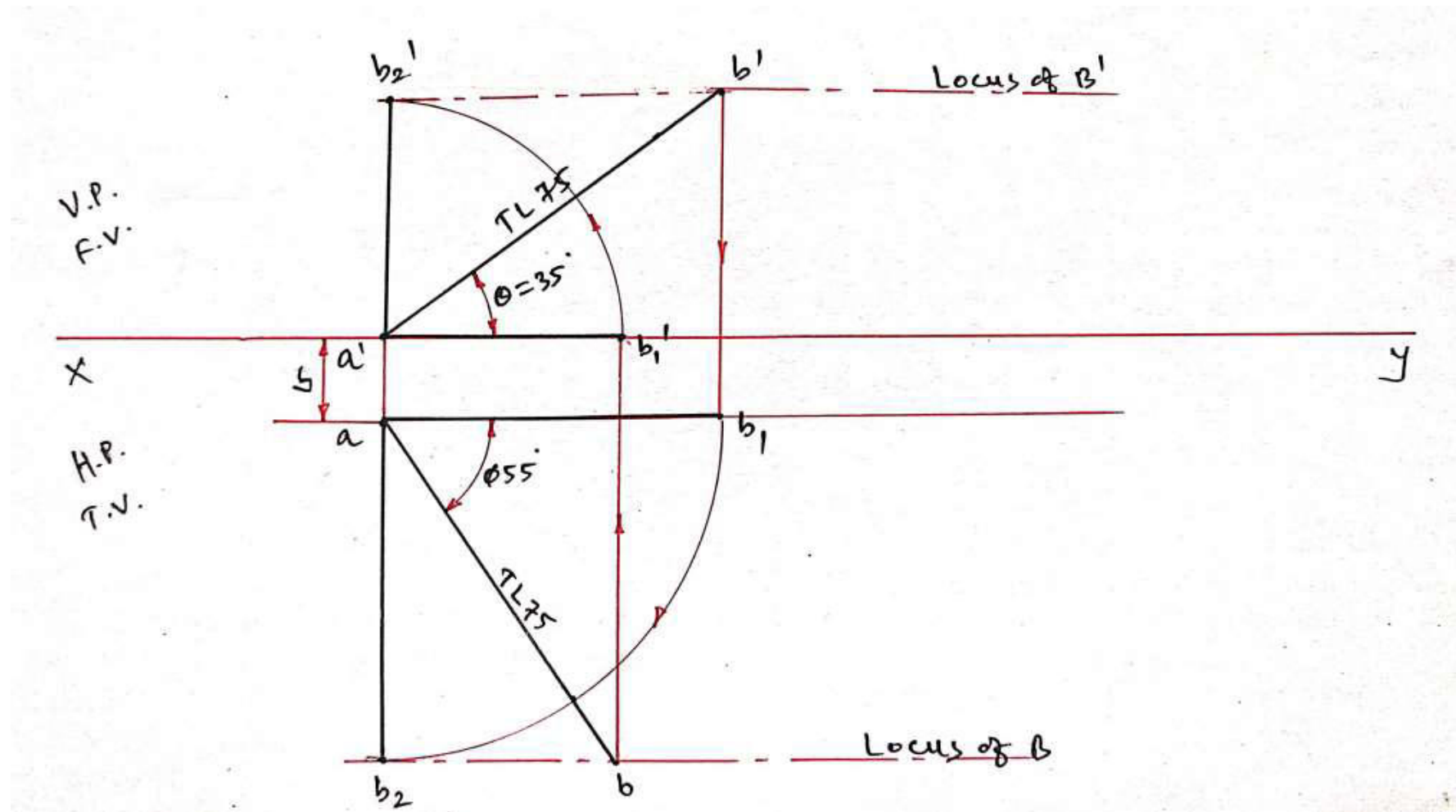


Q.4 (b) OR A square pyramid, base 30 mm side and axis 60 mm long has its base in H.P. all edges of the base are equally inclined to V.P. It is cut by a section plane perpendicular to V.P. and inclined at 45° to the H.P. such that it bisects the axis. Draw its sectional top view and front view and true shape of the section. (04 Marks)



Video Link: <https://youtu.be/2zyRgdDD1QM>







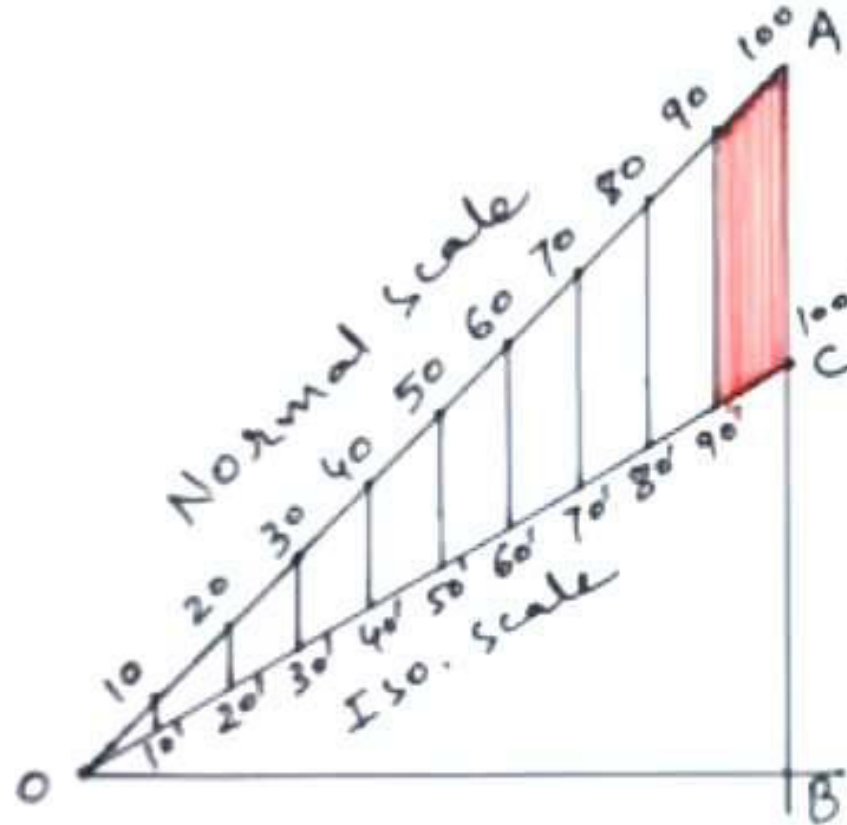
Q.5 (a) Write the advantages of CAD. (03 Marks)

Advantages of CAD.

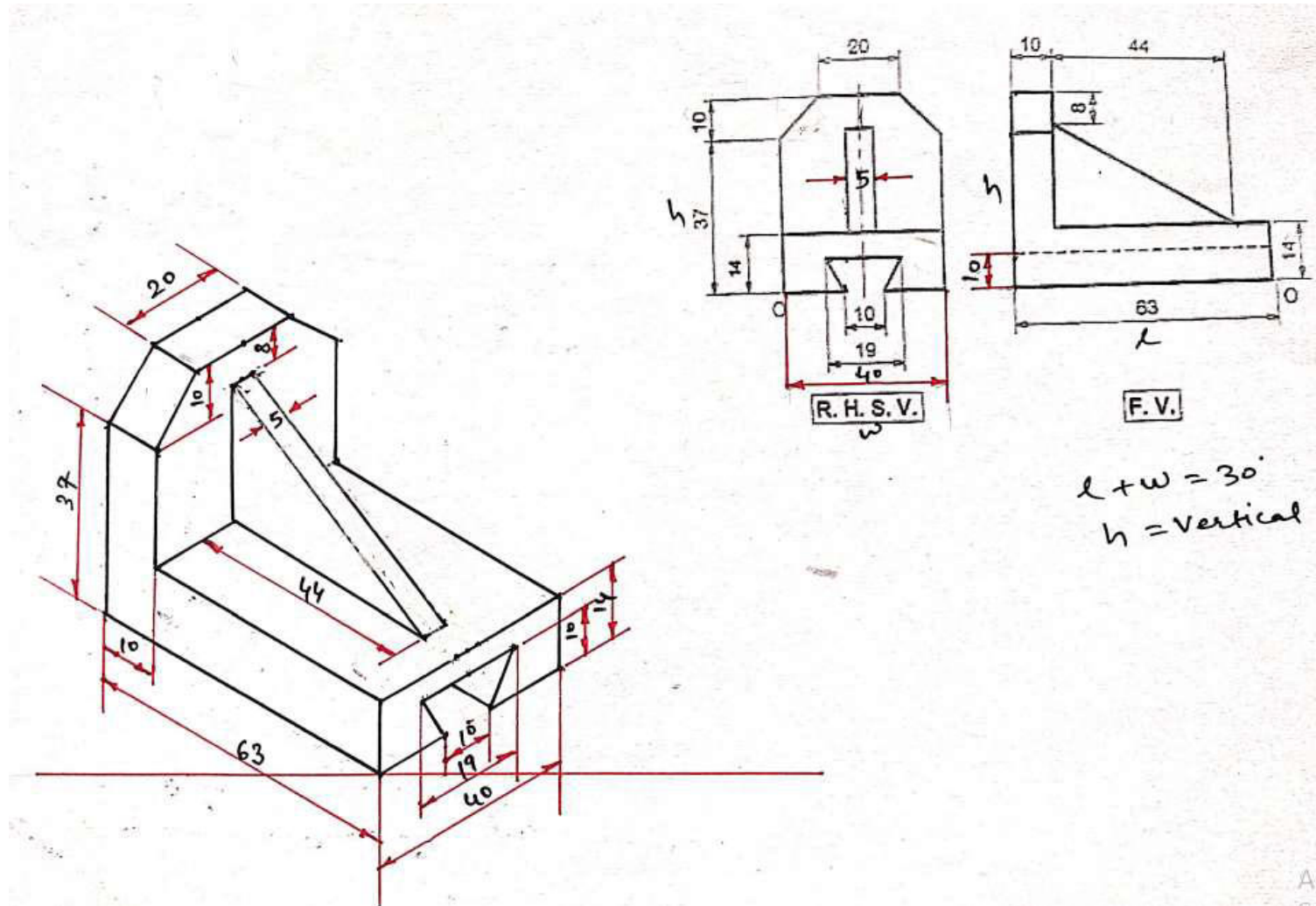
1. With the help of CAD highly accurate and precise drawing can be prepared in less time than manually prepared drawings.
2. Drawing instruments and hand skill is not required.
3. Drawings can be easily saved in computer and it can be modified and copied when necessary.
4. We can print the drawings on printer and plotter when required.
5. No scrap of drawings and no paper wastage.
6. Multicolor display makes visualization more effective.
7. We can increase or decrease the size of drawing to definite scale very easily.
8. Computer aided manufacturing is also possible.



Q.5 (b) Prepare an isometric scale to measure 40 mm and 74 mm. (04 Marks)



Video Link: <https://youtu.be/Bn5WVRldWm0>





Q.5 (a) OR Write various commands used in AutoCAD and their usage. (03 Marks)

Following is the command in draw tool bar.

1. Line : Used to draw the lines
2. Polygon : Used to draw any regular polygon i.e. triangle, square, pentagon, hexagon etc. of any size.
3. Rectangle : Used to draw rectangle of any size.
4. Circle : Used to draw circle of any size.

Following is the command in modify tool bar.

1. Move : Used to move object at required place.
2. Copy : Used to copy the object number of times at required place.
3. Mirror : Used to create mirror image of object.
4. Scale : Used to increase or decrease the size of object.



Q. 5 (b) OR Write the comparison between first angle projection method and third angle projection method.
(04 Marks)

First Angle Projection Method	Third Angle Projection Method
1. Object is placed in first quadrant	1. Object is placed in third quadrant
2. Object is placed between plane and observer	2. Plane is placed between object and observer
3. Plane should not be considered as transparent	3. Plane should be considered as transparent
4. Front view is drawn above reference line XY	4. Front view is drawn below reference line XY
5. Top view is drawn below the front view	5. Top view is drawn above the front view
6. Left hand side view is drawn right side of front view	6. Left hand side view is drawn left side of front view
7. Right hand side view is drawn left side of front view	7. Right hand side view is drawn right side of front view
8. Bottom view is drawn above the front view.	8. Bottom view is drawn below the front view.
9. Rear view can be drawn beside L.H.S.V. or R.H.S.V.	9. Rear view can be drawn beside L.H.S.V. or R.H.S.V.

[Video Link: https://youtu.be/zmAlNodrt1U](https://youtu.be/zmAlNodrt1U)



Q. 5 (c) OR Draw orthographic views of the figure 3 by using first angle projection method:

(i) Full sectional front view

(ii) Top view (07 Marks)

